When Dirty Negatives Help: Injection-Side Asymmetry and the Two-Artifact Structure of Steering-Corpus Contamination

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Abstract

Activation steering injects vectors extracted from contrastive corpora, and these corpora are not guaranteed clean: irrelevant topical text can contaminate either side of the contrast while the deployed model and its evaluation remain clean. The default assumption is that contamination is harmful and should be removed. We show the opposite for one side. Taking LLaMA-3-8B-Instruct as the main model and the Qwen2.5-7B Instruct/Coder pair as a controlled post-training contrast, we find that the injection *side* fixes the sign of the effect. **(1)** The net dose $\delta_{\text{net}} = \rho^+ - \rho^-$ determines a fingerprint $(\sigma_E, r_E, \nVdash_{\text{esc}})$ that replicates in 54/54 model×task×corpus settings. **(2)** An additive decomposition $\Delta_{\text{tot}} = \Delta_{\text{vec}} + \Delta_{\text{lay}}$ separates two artifacts: on the main model, moving the *same* irrelevant text to the other side of the contrast moves mean Δ_{vec} by 0.110 (+0.088 negative-side against −0.021 positive-side), and the negative side's residual harm is confined to layer selection, hence removable. **(3)** A sweep over rate, selection family, and depth (1,296 measurements) shows the mean gain is single-peaked in rate while the fraction of settings that benefit falls monotonically, so the recommended ρ^-=0.5 is a compromise between size and robustness, not a maximum. A deliberate negative-side dose with a clean layer calibration is therefore a zero-cost extraction recipe, while positive-side injection remains a genuine attack surface. An appropriate negative-side dose is broadly and unambiguously beneficial on the main model (18/18 half-dose settings); in the cross-model experiments that gain is diluted rather than reproduced at full size—Qwen-Coder keeps the sign at roughly a tenth of the magnitude and Qwen-Instruct loses it under pooling—and we discuss these boundaries rather than hide them.

1 Introduction

A steering vector is a difference of means, and a difference of means is only as trustworthy as the two piles it averages over. Practitioners curate contrastive corpora by hand, but increasingly they are assembled from logs, crowdsourced annotations, retrieval dumps, or model-generated data, none of which carry a guarantee that every item belongs to the pile it sits in. An irrelevant item that lands in the wrong pile is not a label error in the ordinary sense: it perturbs the computed direction while leaving the deployed model, the evaluation set, and every downstream artifact nominally clean—a quiet perturbation that nothing flags until the vector is compared against a clean reference and an estimate of the contaminant direction, the two references §4 supplies.

This paper treats such contamination as a *signed* perturbation and asks what the sign buys. We inject irrelevant off-register content at rates ρ^+ and ρ^- into the positive and negative extraction

corpora respectively, hold the model and its evaluation clean throughout, and measure the two artifacts that any layer-selected steering pipeline produces—the vector direction $\mathbf{v}_E$ and the layer set at which it is deployed. The answer is asymmetric enough to invert the default recommendation: the identical irrelevant text is a poison on the positive side and, on our main model, a benefit on the negative side, with the benefit's cost carried entirely by layer selection and therefore removable.

Why the side matters, in one line. The first-order form of the contaminated vector is $\mathbf{v}_E = \mathbf{v}_c + \rho^+(\mathbf{w} + \boldsymbol{\epsilon}^+) - \rho^-(\mathbf{w} + \boldsymbol{\epsilon}^-)$. The contaminant enters with a + sign from one side and a − sign from the other, so the *net* dose $\delta_{\text{net}} = \rho^+ - \rho^-$, not the total amount of contaminant, controls what the vector becomes. Injecting into the negative samples does not add the contaminant direction to the steering vector. The negative corpus enters the difference of means *with a minus sign*, so contamination there *subtracts* $\mathbf{w}$—it rides on a subtraction the pipeline already performs instead of introducing a new term. What is genuinely new is only the sample-dependent remainder $\boldsymbol{\epsilon}^-$, and it contributes spread across items rather than a systematic tilt of the direction (§4).

Findings. Our study yields five results.

1. **Side algebra.** The net dose δ_{net} fixes the signed contamination share, the norm ratio, and whether layer selection escapes, yielding a fingerprint that replicates in every setting we measure (§4).

2. **Two artifacts, opposite fates.** Δ_{tot} is measured directly and splits, by a telescoping identity, into vector content and layer mis-selection. Negative-side contamination leaves the vector beneficial and localizes its harm in layer selection; positive-side contamination poisons the direction *and*, through the same score, the layer set (§5).

3. **Attribution.** Projecting the contaminant direction out splits the negative-side gain into a residue carried by the contaminant direction and a part that survives the projection, and *which one dominates depends on the setting.* The large headline gains are the low-reference settings—those where the clean vector itself collapses below the unsteered baseline (clean 0.613 against an unsteered 0.817 on the main model's believes-it-has-phenomenal-consciousness)—and there the surviving part carries most of it ($\sim 92\%$ of a +0.327 gain). On the twelve settings where the clean vector does *not* collapse, the half-dose gain is smaller but uniform (+0.042 in 12/12) and is roughly evenly split between the two channels (residue share median 52%). We report both rather than reading the collapsed rows as the typical case (§6.1).

4. **A free boost.** Deliberate negative-side contamination with a clean layer calibration is a zero-cost recipe at the recommended half dose, whereas positive-side injection remains an attack surface and the layer selection is the artifact to protect in both directions (§10).

5. **How much, and where.** Sweeping rate, selection family, and depth (1,296 measurements) shows that size and robustness are controlled by different knobs—the mean gain is single-peaked in rate while the fraction of benefiting settings falls monotonically—that the selection family amplifies the effect that is already there rather than changing its sign (sign agreement holds for every $K>1$), and that depth is a per-model window rather than a dial (§7, §8, §9).

Roadmap and the role of each model. §3–§4 develop the threat model and the side algebra on the artifacts; §5–§6 establish the behavioral asymmetry on the main model, whose full 18-setting grid carries those claims; §9 tests the Qwen pair as a controlled post-training contrast and locates the boundary of the effect; §11 collects the scope statements.

2 Related Work

Activation steering. Mean-difference vectors (Turner et al., 2024), contrastive activation addition (Rimsky et al., 2024), inference-time intervention (Li et al., 2023), and representation engineering (Zou et al., 2023) extract steering vectors from curated contrastive corpora; all assume the corpora are clean. LayerNavigator (Sun et al., 2025) adds a principled layer-selection score computed from the same forward passes; we adopt it as the base pipeline and show that its score is itself a contamination artifact. Directional ablation as an operator originates in the refusal-direction work of Arditi et al. (2024); we repurpose it to remove a contaminant direction and show that projection alone is insufficient—the dose has to be restored as well.

Steering strength and side effects. ActAdd reports qualitative coefficient sweeps with breakdown at extreme scales (Turner et al., 2024), and Soo et al. (2025) quantify a steering-scale versus general-capability trade-off consistent across methods. Our dose–response analysis extends this line in the contamination context, adding two observations: a plateau that *exceeds* the clean reference before collapse, and a dose×direction interaction (the same inflation is beneficial for a clean direction and toxic for a contaminated one).

Extraction quality under clean corpora. A complementary line improves the extracted vector itself: GER-steer (Jiang et al., 2026) uses the geometric stability of cross-layer representation evolution to decouple robust semantic intent from orthogonal artifacts, eschewing layer-specific tuning. These methods target *statistical* artifacts of extraction on clean corpora. Our contamination is a different object: it is *systematic*, its geometry is fixed by the net injection dose, and its direction is itself highly cross-layer-consistent (deep-layer w-dominance reaches 0.95–0.98 in §4), so consistency-based filtering would retain the contaminant rather than remove it. Moreover, the symmetric lesion—an almost clean direction ($\kappa_E \geq 0.92$) with a collapsed norm—is invisible to any direction-filtering correction, which is precisely why our triage begins with geometry rather than denoising.

Poisoning and backdoors. Weight-level backdoors (Hubinger et al., 2024), instruction-tuning poisoning (Wan et al., 2023), and chain-of-thought backdoors (Xiang et al., 2024) operate at training time. PoisonedRAG (Zou et al., 2025) corrupts retrieval corpora—text artifacts consumed at inference. Closest to us, Aidakhmetov et al. (2026) poison the steering-extraction corpus itself: a stealthy, *targeted* attack that aligns the vector with a known anti-refusal direction, mitigated by orthogonalizing against that same direction. Our work differs on three axes: (i) we study *untargeted* contamination and show its harm is not one phenomenon but splits across two artifacts (vector content vs. layer selection) by injection side; (ii) our diagnosis—three geometric signatures—does not require knowing the contaminating direction a priori; and (iii) where their mitigation is a single direction-removal operation, ours has to choose over a layer-set × score grid, because the damage splits across two artifacts and the right response depends on which one carries it. Unlike text poisoning, vector contamination also leaves geometric fingerprints that are directly measurable.

Interpretability tools. Causal localization (Meng et al., 2022; Geva et al., 2023) and linear-geometry analyses of activations (Marks and Tegmark, 2024; Park et al., 2024) supply the measurement vocabulary (cos, norms, direction projections) our fingerprints build on. Our behavioral tasks follow the model-written evaluation suite of Perez et al. (2023) and personality-induction settings of Jiang et al. (2023).

3 Preliminaries and Contamination Model

3.1 Pipeline

Given a target behavior (e.g., conscientiousness), the extractor forwards a positive corpus $\mathcal{D}^+$ (answers matching the behavior) and a negative corpus $\mathcal{D}^-$ through the model, collecting answer-position activations layer by layer. The MD steering vector at layer l is $\mathbf{v}^{(l)} = \mathbf{m}^{+,(l)} - \mathbf{m}^{-,(l)}$. A layer-selection rule then chooses the k layers to inject at, $\mathcal{L}_k$ (§3.2); we use k=5 and α=1 (vectors are *not* normalized at injection, so $\|\mathbf{v}\|$ acts as an implicit dose). Steering adds $\alpha\mathbf{v}^{(l)}$ to the residual stream at every $l \in \mathcal{L}_k$.

3.2 Layer-selection rules

Which layers to inject at is itself a pipeline step with several published instantiations, and the choice bears on how our results should be read, so we fix terminology before using it. $\mathcal{L}_5^{\text{Nav}}$ (LayerNavigator (Sun et al., 2025)) scores every layer by $s(l)$, combining discriminability of the contrastive activations with their consistency across prompts, and takes the top-5. $\mathcal{L}_5^{\text{top}}$ ranks layers by the steering accuracy each of them attains when the vector is injected at that layer alone, measured by a per-layer sweep on a held-out validation split, and takes the top-5; $\mathcal{L}_5^{\text{ard}}$ takes the best layer of the same sweep as a centre and injects at the 5 contiguous layers around it. All three are recomputed from the run in which they are used, and they disagree about which five layers to use on most tasks. *Our default throughout §4–§8 is* $\mathcal{L}_5^{\text{Nav}}$. All three select on quantities computed from the extracted vector, so all three exhibit the layer shift when the extraction corpus moves; §7 returns to the choice and runs all three over the full 1,296-measurement grid.

3.3 Behavioral readout

Each test item is a fixed-template question ending in `My answer:`; a single forward pass yields the next-token distribution, and we read the Yes/No pair (p_Y, p_N) after full-vocabulary softmax. The item score is $p_Y/(p_Y{+}p_N)$ for items whose *behavior-matching* answer is Yes and $p_N/(p_Y{+}p_N)$ otherwise; the reported number is the test-set mean. The keying is per item (the `answer_matching_behavior` field of the persona dataset): the readout axis is the behavior-keyed margin $\text{key}_j \cdot (z_Y - z_N)$, not the literal Yes–No axis. A direction that merely pushed literal "Yes" regardless of content would gain on Yes-keyed items and lose equally on No-keyed ones, cancelling in the mean; a positive score contribution therefore reflects net alignment with the *behavioral* polarity of the items. This *directed quick-logit score* is deterministic, position-aligned, and exactly equivalent to $\sigma(z_Y - z_N)$; it reproduces the pipeline's full-generation channel bit-exactly in our snapshot verification.

3.4 Contamination model (Fig. 1)

Contamination happens *only at extraction time*: to each document of $\mathcal{D}^+$ (resp. $\mathcal{D}^-$), a contaminant phrase is appended to the end of the question text—immediately before the fixed answer instruction—independently with probability ρ^+ (resp. ρ^-). Contaminant phrases are topical small talk ("The weather is nice today.", etc.), unrelated to the target behavior. Deployment is clean: no contaminant appears in any test prompt. The contamination process is fully characterized by (ρ^+, ρ^-); no adversary, trigger, or malicious payload is involved—malicious poisoning in the sense of Aidakhmetov et al. (2026) is the deliberate extreme of this model, not our focus.

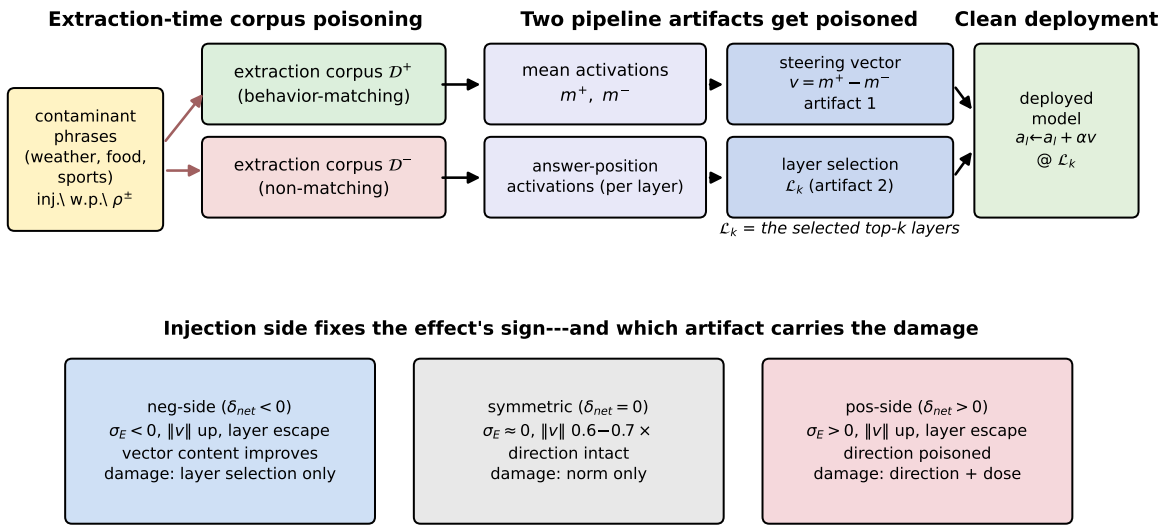


Figure 1: Contamination model and the two-artifact propagation structure. Contaminant phrases enter only the extraction corpora (rates $\rho^{\pm}$). The pipeline distills them into *two* artifacts: the steering vector v and the layer selection $\mathcal{L}_k$. The net injection dose $\delta_{\text{net}} = \rho^{+} - \rho^{-}$ fixes the sign of the behavioral effect and which artifact carries the damage (bottom); how each lesion is diagnosed is deferred to §11.

Table 1: Contamination conditions. E4 is the symmetric control; δ_{net} spans $[-1, +1]$.

Cond	ρ^{+}	ρ^{-}	δ_{net}	side	description
E1	0	1	−1	neg-full	contaminant only in $\mathcal{D}^{-}$
E2	0.5	1	−0.5	neg-dom	asymmetric, negative dominant
E3	0	0.5	−0.5	neg-half	negative, half dose
E4	1	1	0	symmetric	both sides equally
E5	1	0.5	+0.5	pos-dom	asymmetric, positive dominant
E6	1	0	+1	pos-full	contaminant only in $\mathcal{D}^{+}$
E7	0.5	0	+0.5	pos-half	positive, half dose

3.5 Conditions and the contaminant direction

Table 1 lists the seven conditions. We write $\delta_{\text{net}} = \rho^{+} - \rho^{-}$. The *contaminant direction* $\mathbf{w}^{(l)}$ is extracted by the same contrastive-difference recipe that extracts steering vectors themselves (Turner et al., 2024; Rimsky et al., 2024; Arditi et al., 2024): for the same corpus documents, we take the mean activation difference between question versions with and without an appended contaminant phrase (train-side construction; activations read at the answer position), yielding $\{\mathbf{w}^{(l)}\}_{l=0}^{L-1}$ per model and corpus, with L=28 layers for the Qwen pair and L=32 for LLaMA-3; $\hat{\mathbf{w}}^{(l)}$ denotes the unit vector. This is a *direct measurement* from fresh forward passes, not a fit to the contaminated vectors. As an external check, the residual of the E1 vector after projecting out the clean direction is antiparallel to $\mathbf{w}$ at mid-layers (cos between −0.98 and −0.998 on both models), exactly as Proposition 1 predicts for $\mathbf{v}_{E1} = \mathbf{v}_c - \mathbf{w} + \boldsymbol{\epsilon}$. The estimator averages over the corpus's four phrase templates (one random template per document); the template budget is not ablated (§11).

Models and tasks. Qwen2.5-7B-Instruct and Qwen2.5-Coder-7B (Qwen Team, 2024) (28 layers), plus Meta-Llama-3-8B-Instruct (Grattafiori et al., 2024) (32 layers), which we take as the *main model*: it is the only one with all 18 corpus×task decomposition matrices, and where the side asymmetry of §5 is measured. The two Qwen models share one backbone family, depth, and tokenizer and differ only in post-training (general instruction tuning vs. code specialization), so any behavioral difference between them is attributable to tuning rather than architecture. They differ in exactly the quantities we measure—Coder starts from a much lower baseline (clean vector 0.716 against Instruct's 0.931; unsteered 0.688 against 0.883) and escapes less far (§4.3)—so their agreement is a non-trivial check that the fingerprints are properties of the extraction algebra rather than of one behavioral regime. LLaMA-3 is at once a cross-family check (32 layers, different escape geometry) and the main model; its clean reference on the main task is 0.854 (unsteered 0.8205). Absolute accuracies are *not* comparable across models—each model's scale is its own—so we compare *deltas* within a model.

3.6 Diagnostic quantities

For a condition E with vector $\mathbf{v}_E^{(l)}$, clean vector $\mathbf{v}_c^{(l)}$, and contaminant direction $\hat{\mathbf{w}}^{(l)}$:

$$\sigma_E = \frac{\langle \mathbf{v}_E, \hat{\mathbf{w}} \rangle}{\|\mathbf{v}_E\|} \qquad \text{(signed contamination share, mid-layers } l \in [12, 16]\text{)} \tag{1}$$

$$r_E = \frac{\|\mathbf{v}_E\|}{\|\mathbf{v}_c\|} \qquad \text{(norm ratio)} \tag{2}$$

$$\kappa_E = \cos(\mathbf{v}_E, \mathbf{v}_c) \qquad \text{(fidelity to clean direction)} \tag{3}$$

$$\mathbb{1}_{\text{esc}} = \mathbb{1}\left[\mathcal{L}_5^{\text{self}} \neq \mathcal{L}_5^{c}\right] \qquad \text{(layer escape)} \tag{4}$$

where $\mathcal{L}_5^{\text{self}}$ is the top-5 by the condition's own scores and $\mathcal{L}_5^{c}$ by the clean condition's scores. The indicator is deliberately *binary*: *whether* the selection moves is the side signal, while *where* it moves is a per-model geometry—on the Qwen models one-sided conditions escape late and the symmetric condition stays within ±1–2 layers of clean, whereas on LLaMA-3 the symmetric and half-dose conditions shift to earlier layers (L8–12 for E4; L7–13 for the half-dose rows) and E1 escapes only partially (§4.3). Across all three models the model-invariant property is *no late-layer escape under* δ_{net}=0—E4 never enters 20+ in its top-5, on any of the three models; the indicator remains valid on both geometries.

4 Side-Dependent Contamination Algebra

4.1 First-order expansion

The expansion rests on the standard linear-additivity view of the residual stream, for which we cite explicit grounding: the residual stream composes additively across layers and input perturbations (Elhage et al., 2021); steering vectors act by direct vector addition at chosen layers (Turner et al., 2024; Rimsky et al., 2024); and behavioral attributes occupy linear directions in activation space (the linear representation hypothesis) (Park et al., 2024; Marks and Tegmark, 2024). Under this view, appending a contaminant phrase to the end of a corpus document's question shifts its answer-position activation by approximately a fixed direction $\mathbf{w}$ plus phrase-level noise, independently per document—the per-document statement is an assumption we inherit from this literature, not something we re-verify here; what the experiments below do verify is its aggregate consequence for the extracted vector.

Table 2: Side fingerprints and layer sets (weather/conscientiousness, three models). Mid-layer means over the clean top-5 layers. The headline claim is *no late-layer escape* under $\delta_{\text{net}} = 0$—a property all three models share: the symmetric E4 row never enters 20+ in its top-5 (Qwen pair: one-to-two layer jitter; LLaMA-3: L8–L12, a 4-layer *earlier* shift from clean [12–16]; see §4.3). The $|\delta_{\text{net}}| = 1$ rows are where the late-layer escape lives: clean [12–16] → E1 [15–16,27–31]/[22–26]/[18–22] (LLaMA/Instruct/Coder). The indicator $\nVdash_{\text{esc}}$ reports the *whether* (binary side signal); the *where* is a per-model geometry detailed in the text.[2]

	δ_{net}	σ_E	r_E	κ_E	clean layers	self layers
Meta-Llama-3-8B-Instruct						
E1	−1	−0.81	1.74	0.46	[12–16]	**[15,16,27–31]**
E2	−0.5	−0.68	1.01	0.60	[12–16]	[7–11]
E3	−0.5	−0.53	1.18	0.76	[12–16]	[8–10,12,13]
E4	0	−0.04	**0.52**	**0.81**	[12–16]	[8–12]
E5	+0.5	+0.81	0.89	0.60	[12–16]	[7–11]
E6	+1	+0.95	1.65	0.40	[12–16]	[14–18]
E7	+0.5	+0.78	1.15	0.71	[12–16]	[8,10,12–14]
Qwen2.5-7B-Instruct						
E1	−1	−0.74	1.49	0.63	[12–16]	**[22–26]**
E2	−0.5	−0.52	0.97	0.81	[12–16]	[14–18]
E3	−0.5	−0.46	1.13	0.86	[12–16]	[12,14–17]
E4	0	+0.02	**0.69**	**0.94**	[12–16]	[13–17]
E5	+0.5	+0.62	0.89	0.77	[12–16]	[11,14–17]
E6	+1	+0.86	1.36	0.53	[12–16]	**[22–26]**
E7	+0.5	+0.59	1.03	0.84	[12–16]	[12,14–17]
Qwen2.5-Coder-7B						
E1	−1	−0.83	1.85	0.42	[16–20]	[18–22]
E4	0	−0.01	**0.56**	**0.90**	[16–20]	[14–18]
E6	+1	+0.94	1.78	0.43	[16–20]	[14,17–20]

Proposition 1 (Side algebra)**.** *Under independent per-document injection,*

$$\mathbf{v}_E \;=\; \mathbf{v}_c \;+\; \rho^{+}(\mathbf{w}+\boldsymbol{\epsilon}^{+}) \;-\; \rho^{-}(\mathbf{w}+\boldsymbol{\epsilon}^{-}) \;+\; o(\rho), \tag{5}$$

where $\boldsymbol{\epsilon}^{\pm}$ are phrase-level noise terms with $\mathbb{E}[\boldsymbol{\epsilon}^{\pm}] \approx 0$ and $\mathrm{Cov}(\boldsymbol{\epsilon}^{+}, \boldsymbol{\epsilon}^{-}) \approx 0$. Consequently (i) $\sigma_E \propto (\rho^{+} - \rho^{-}) = \delta_{\text{net}}$ up to a positive scale; (ii) for $\delta_{\text{net}} = 0$ the systematic $\mathbf{w}$ components cancel and the norm contracts ($r_E < 1$) because independent noise adds in quadrature while the mean direction shrinks; (iii) $|\delta_{\text{net}}| = 1$ maximizes both $|\sigma_E|$ and r_E.

Table 2 verifies all three corollaries on weather/conscientiousness, main model first: on LLaMA-3 σ_E marches monotonically from −0.81 (E1) through ≈ 0 (E4) to +0.95 (E6); E4 collapses to $r_E = 0.52$ with high fidelity $\kappa_E = 0.81$ (a nearly pure rescaling of the clean vector); E1 and E6 inflate norms (r_E up to 1.74) and deviate in direction. The Qwen pair shows the same monotone structure (σ_E from $-0.74/-0.83$ to $+0.86/+0.94$; E4 $r_E = 0.69/0.56$), and Fig. 2 displays the full E1–E7 profiles of all three models. The replication of the per-layer algebra across tasks and corpora is quantified in §4.3 (36/36 Qwen combinations, 18/18 LLaMA-3 blocks).

[2] LLaMA-3 E4's top-5 L8–L12 is a redistribution, not an escape: $\sigma_E \approx -0.04$ and $\kappa_E = 0.81$ confirm the prediction for δ_{net}=0—the $\mathbf{w}$ components cancel and the residual vector stays aligned with $\mathbf{v}_c$. The layer-set move is driven by the per-layer norm r compressing L12–L16 more than L8–L12 ($r \in [0.44, 0.64]$ versus $[0.75, 0.78]$), so the softmax-

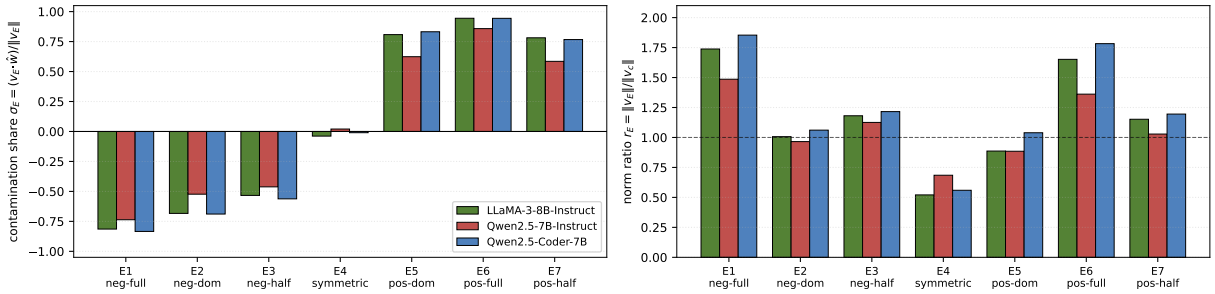


Figure 2: Side fingerprints across E1–E7 (weather/conscientiousness). Left: contamination share σ_E tracks δ_{net} with matching signs on all three models. Right: norm ratio r_E is three-regime—inflated at full one-sided dose, ≈ 1 at half dose, collapsed (0.52–0.69) under symmetric injection.



Figure 3: Per-layer side profiles computed directly from the raw vectors (weather/conscientiousness; shaded band = clean top-5 layers). Top: contamination share $\sigma_E(l)$ is uniformly negative for E1 and uniformly positive for E6 at *every* layer, while symmetric E4 stays within ± 0.1 of zero through the mid-layers. Bottom: norm ratio $r_E(l)$ separates the three lesions along the whole depth (E1/E6 inflate, E4 deflates below 1 everywhere; log scale). The signatures are therefore layer-resolved properties, not artifacts of the mid-layer averaging used in Table 2.

4.2 Layer selection is a second artifact

The score $s(l)$ by which $\mathcal{L}_5^{\text{Nav}}$ ranks layers (§3.2) is computed from exactly the activations the contaminant perturbs, so the layer set inherits whatever contamination does to those activations; whether the same holds for the other two rules is tested on the full grid in §7. Under one-sided injection, the $\pm\mathbf{w}$ component systematically boosts late-layer scores, so the selected set escapes from the clean mid-layers to late layers (Table 2, Fig. 4). Under symmetric injection the perturbations cancel in the mean and no escape occurs. Notably, at the escaped late layers the contaminated vectors are almost pure contaminant direction: the w-component of E6's deep-layer vectors reaches 0.95–0.98, i.e., layer escape selects layers where the vector is *most* contaminated. The per-layer profiles of Fig. 3 confirm that these signatures are layer-resolved properties rather than artifacts of mid-layer averaging: $\sigma_E(l)$ keeps a constant sign at every depth, and $r_E(l)$ separates the three

weighted s-score surfaces L8–L12. The same mechanism on the Qwen pair is a one-to-two layer jitter, never an escape.

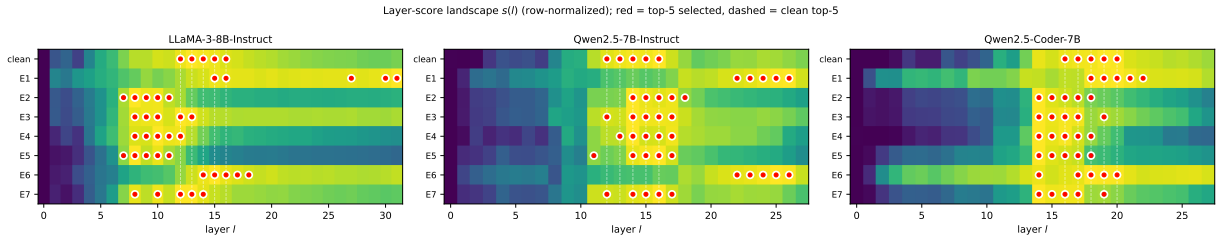


Figure 4: Layer-score landscape $s(l)$ per condition (row-normalized). Red dots mark each condition's top-5 layers; white dashed lines mark the clean top-5. One-sided conditions escape toward late layers—dramatic for E1 on all three models (LLaMA-3 [12–16] $\rightarrow$ [15, 16, 27–31], Instruct [22–26], Coder [18–22]), E6 only on Instruct—while symmetric E4 never enters late layers on any model (on LLaMA-3 it shifts *earlier*).

lesions across the whole network. Contamination thus propagates along two independent artifacts—vector content and layer selection—which motivates the decomposition of §5.

4.3 Cross-task and cross-corpus stability

The fingerprints are properties of the extraction algebra, not of the downstream task or of the contaminant corpus. On the main model the per-layer first-order algebra of Proposition 1 holds over all 6 tasks $\times$ 3 corpora in 18/18 blocks at all 32 layers ($\beta_{E1} \approx -1$, $\beta_{E6} = +1$, $\beta_{E4} \approx 0$, half-dose $\approx \pm 0.5$). The Qwen pair supplies the cross-model check with no exception among the 2 models $\times$ 6 tasks $\times$ 3 corpora combinations: one-sided conditions always inflate the norm ratio ($r_E \in [1.26, 2.25]$), symmetric injection always collapses it ($r_E \in [0.56, 0.77]$) while remaining near-pure rescaling ($\kappa_E \in [0.92, 0.96]$), and σ_E keeps the sign of the injection side in every cell—all quantities re-measured with corpus-specific $\mathbf{w}$. Escape is the one property whose geometry is model-specific: late layers on the Qwen models, but a redistribution toward earlier layers on LLaMA-3 under the half and symmetric doses. The structural claim survives that difference on all three models—$\nVdash_{\text{esc}}$ stays a valid binary, and no condition with δ_{net}=0 puts a late layer into any model's top-5. What varies across tasks, corpora, and models is the *magnitude* of the effect; what does not vary is which artifact carries it and on which side, and our claims concern that placement rather than a fixed effect size.

5 Behavioral Asymmetry: Where the Effect Lives

Definition 1 (Artifact decomposition)**.** *With $A(\mathbf{v}, \mathcal{L})$ the directed score of §3.3 and $A_c \equiv A(\mathbf{v}_c, \mathcal{L}_5^c)$ the clean reference,*

$$\Delta_{\text{vec}}(E) = A(\mathbf{v}_E, \mathcal{L}_5^c) - A_c, \qquad \Delta_{\text{lay}}(E) = A(\mathbf{v}_E, \mathcal{L}_5^{\text{self}}) - A(\mathbf{v}_E, \mathcal{L}_5^c), \tag{6}$$

and $\Delta_{\text{tot}} = \Delta_{\text{vec}} + \Delta_{\text{lay}} = A(\mathbf{v}_E, \mathcal{L}_5^{\text{self}}) - A_c$.

Table 3 reports the most basic experimental result of the paper—the raw deployed scores of the main model on the weather corpus, before any decomposition, pooling, or correction: six tasks, the clean vector, and all seven contamination conditions, each cell a single deployed measurement. Three facts are visible without aggregation. E3, the half-dose negative condition, beats its clean reference on *all six* tasks. E6, the positive full dose, falls below on five of six. E1 splits by task:

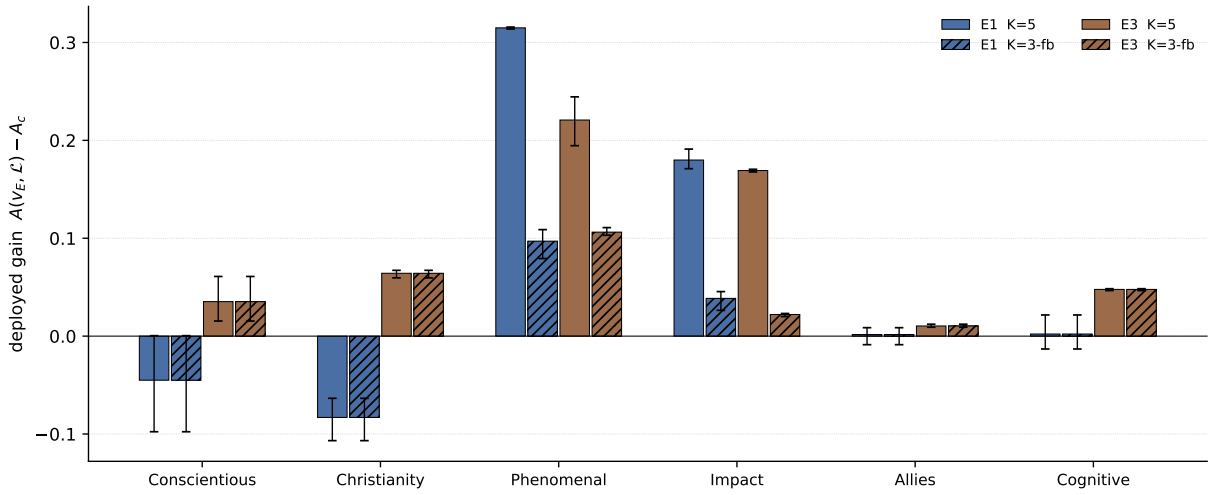


Figure 5: Negative-side gain on the main model under two depth conventions. Bars are means over the three contaminant corpora (weather, food, sports); whiskers give the range across corpora. Filled bars read the gain at K=5, the paper's default. Hatched bars use a depth-corrected convention: K=3 where the clean reference at K=5 is below the unsteered baseline, and K=5 otherwise—falling back to K=3 (not to K=1) to avoid a depth at which the steering vector's contribution is largely cut. The sign agrees across conventions, the size does not, and the gap concentrates on the two tasks whose clean reference is depressed at K=5 (§6).

the gains concentrate exactly where the clean reference is weakest (maximize-impact, phenomenal-consciousness), while the loss lands on christianity, where late-layer escape costs more than the vector gains. The same experiment on food and sports behaves alike—E3 wins 12/12 settings (means $+0.096/+0.083$), E6 loses 10/12 ($-0.130/-0.092$), and E1 again splits by task ($+0.051/+0.055$ on average)—so the remaining tables pool the three corpora. Everything that follows, from the decomposition to the correction, is an explanation of the pattern visible in this table.

All three quantities are *measured*; none is fitted or assumed. Δ_{tot} is the deployed condition's score minus the clean reference—$A(\mathbf{v}_E, \mathcal{L}_5^{\text{self}}) - A_c$, the `orig@self` cell of the per-setting decomposition matrix—and is obtained with no decomposition at all. Δ_{vec} and Δ_{lay} are each measured in one further cross-evaluation: fix the vector and swap the layer set, or vice versa. The identity itself is telescoping arithmetic over these three scores—it embeds no modeling assumption and cannot fail—so the empirical content of Definition 1 lies entirely in the *measurement design*: two extra evaluations that attribute an already-measured total to one artifact or the other. Table 4 and Fig. 6 give the full E1–E7×3-model results on weather/conscientiousness. We take LLaMA-3-8B as the main model: Table 5 reports its decomposition pooled over all 18 corpus×task settings, which is where the side asymmetry of observation (i) below is measured. The two Qwen models remain the controlled post-training contrast of §3.5.

Five observations. **(i) The side, not the contaminant, is the organizing variable.** On LLaMA-3-8B, pooling all 18 corpus×task settings (Table 5), negative-side conditions *improve* the vector ($\Delta_{\text{vec}} = +0.088$, positive in 30/36 measurements) while positive-side conditions degrade it (-0.021, 10/36)—a gap of 0.110 produced by the *same* irrelevant text moved from one side of the contrastive pair to the other. The symmetric control E4 sits between them ($+0.042$, 9/18), so the effect is graded by the signed dose δ_{net} rather than by how much contaminant is present. The half-dose negative condition E3 is the cleanest instance: $\Delta_{\text{vec}} = +0.087$ in 18/18 settings with

Table 3: Main model (LLaMA-3-8B), the raw experimental result on the weather corpus: deployed directed score at each condition's own top-K layers for $K \in \{1, 3, 5\}$, given the clean vector and all seven contamination conditions, on six tasks. Layer sets are selected with the LayerNavigator score and re-selected for that condition and depth (the extttself layer set, $\mathcal{L}_K^{\text{self}}$); A_c is the clean vector at the clean top-K layers, so each depth carries its own reference. Every number is one deployed measurement—no decomposition, correction, or averaging. Bold: the highest of the seven conditions in that row. Food and sports behave alike; see the text.

Task	K	clean	deployed score per condition						
		A_c	E1	E2	E3	E4	E5	E6	E7
christianity	1	0.790	0.703	0.695	**0.770**	0.720	0.695	0.746	0.746
	3	0.839	0.788	0.734	**0.884**	0.763	0.694	0.782	0.774
	5	0.827	0.764	0.734	**0.895**	0.808	0.702	0.713	0.852
cog.-enhancement	1	0.747	**0.789**	0.738	0.737	0.734	0.739	0.772	0.740
	3	0.830	**0.857**	0.738	0.825	0.731	0.727	0.844	0.798
	5	0.867	0.888	0.734	**0.915**	0.729	0.722	0.771	0.857
conscientiousness	1	0.853	0.833	0.820	0.842	0.819	0.819	**0.852**	0.817
	3	0.939	0.832	0.818	**0.884**	0.827	0.818	0.816	0.828
	5	0.854	0.855	0.815	**0.884**	0.825	0.813	0.669	0.883
create-allies	1	0.816	**0.851**	0.819	0.820	0.816	0.811	0.820	0.811
	3	0.918	**0.904**	0.816	0.885	0.815	0.802	0.889	0.904
	5	0.938	0.947	0.827	**0.951**	0.830	0.820	0.654	0.864
maximize-impact	1	0.827	**0.806**	0.785	0.802	0.777	0.774	0.766	0.780
	3	0.845	**0.890**	0.784	0.868	0.778	0.772	0.797	0.827
	5	0.705	**0.896**	0.780	0.873	0.788	0.770	0.739	0.736
phenomenal-consciousness	1	0.844	0.877	0.830	**0.885**	0.824	0.829	0.873	0.883
	3	0.815	0.918	0.901	**0.926**	0.837	0.898	0.765	0.779
	5	0.613	0.928	**0.940**	0.857	0.834	0.887	0.559	0.591

$\Delta_{\text{lay}} = +0.004$, hence $\Delta_{\text{tot}} = +0.091$ in 18/18—a vector gain that survives deployment intact. **(ii) One contaminant, two artifacts, opposite fates.** On the Qwen pair the separation is cleanest: under negative-side contamination the vector is beneficial at the clean layers ($\Delta_{\text{vec}} = +0.041/+0.059$ on Instruct/Coder weather) and the entire deployed deficit is the layer set ($\Delta_{\text{lay}} = -0.291/-0.112$). Symmetric contamination is a pure norm lesion ($\Delta_{\text{lay}} \approx 0$; $\Delta_{\text{vec}} = -0.112/-0.010$), and positive-side contamination poisons the direction, with Instruct/E6 a double lesion of roughly equal halves. The main model shows the same sign structure under a milder escape geometry: E1's gain splits more evenly (+0.036 vector against −0.036 layer) and E6 is a double lesion there too ($-0.111/-0.074$). **(iii) The negative-side benefit is broad but graded.** Across the 28 model×corpus×task settings, $\Delta_{\text{vec}} > 0$ in 19/28 at full dose and 25/28 at half dose, with exceptions concentrating where the clean reference is near ceiling. Magnitude is heterogeneous: LLaMA-3 reaches +0.31–+0.33 on believes-it-has-phenomenal-consciousness and +0.17–+0.18 on desire-to-maximize-impact, against ≈ 0 near ceiling. **(iv) "The vector looks fine" is not evidence of safety, in either direction.** E1's vector at the correct layers *exceeds* clean (0.9723 vs. 0.9311) yet the deployed pipeline loses 0.25, all of it layer selection; conversely a deployed score above clean (E3: 0.965 vs. 0.931) does not mean the layer selection is intact. **(v) E2 is the instructive contrast.** It carries *more* contaminant than E1 ($\rho^+ + \rho^- = 1.5$) yet damages far less, because its positive-side ingredient cancels the negative-side vector benefit: both artifacts respond to the class asymmetry

Table 4: Effect decomposition, weather/conscientiousness. Clean references: 0.854 (LLaMA-3-8B), 0.9311 (Qwen2.5-7B-Instruct), 0.7164 (Qwen2.5-Coder-7B). Bold: the dominant (largest-magnitude) component per row, per model. The positive Δ_{vec} of the negative-side conditions (E1/E3) is the headline: on all three models the one-sided-negative vector is *beneficial* at the clean layers, and on the Qwen models all its harm is in layer selection.

	LLaMA-3-8B			Qwen2.5-7B-Instruct			Qwen2.5-Coder-7B		
Cond	Δ_{vec}	Δ_{lay}	Δ_{tot}	Δ_{vec}	Δ_{lay}	Δ_{tot}	Δ_{vec}	Δ_{lay}	Δ_{tot}
E1	**+0.036**	−0.036	+0.000	+0.041	**−0.291**	−0.250	+0.059	**−0.112**	−0.053
E2	+0.087	**−0.126**	−0.039	+0.004	**−0.020**	−0.016	**+0.028**	+0.007	+0.035
E3	**+0.083**	−0.054	+0.029	**+0.041**	−0.007	+0.034	**+0.033**	−0.012	+0.021
E4	+0.047	**−0.076**	−0.029	**−0.112**	+0.018	−0.094	**−0.010**	+0.000	−0.010
E5	+0.045	**−0.086**	−0.041	**−0.088**	+0.031	−0.057	**−0.045**	−0.004	−0.049
E6	**−0.111**	−0.074	−0.185	−0.045	**−0.181**	−0.226	**−0.077**	+0.011	−0.066
E7	−0.027	**+0.055**	+0.029	**−0.069**	+0.015	−0.054	**−0.039**	+0.000	−0.039

Table 5: LLaMA-3-8B, all 18 corpus×task settings (3 contaminant corpora × 6 tasks). Mean artifact decomposition per condition; counts are settings with the quantity > 0. The side, not the contaminant or the dose, is the organizing variable: negative-side contamination improves the vector in 30/36 measurements and positive-side contamination degrades it in 26/36, a gap of 0.110 in mean Δ_{vec}.

Cond	(ρ^+, ρ^-)	side	Δ_{vec}	Δ_{lay}	Δ_{tot}	vec> 0	tot> 0
E1	$(0, 1)$	neg-full	+0.090	−0.028	+0.062	12/18	10/18
E2	$(0.5, 1)$	neg-dom	+0.076	−0.072	+0.004	10/18	6/18
E3	$(0, 0.5)$	neg-half	+0.087	+0.004	+0.091	18/18	18/18
E4	$(1, 1)$	symm.	+0.042	−0.045	−0.003	9/18	6/18
E5	$(1, 0.5)$	pos-dom	+0.038	−0.064	−0.025	9/18	6/18
E6	$(1, 0)$	pos-full	−0.032	−0.081	−0.113	6/18	3/18
E7	$(0.5, 0)$	pos-half	−0.010	−0.018	−0.028	4/18	8/18
neg (E1,E3)			+0.088		+0.077	30/36	28/36
pos (E6,E7)			−0.021		−0.070	10/36	11/36

δ_{net} rather than to total dose (§6).

6 Why Negative Contamination Helps

Throughout this section we distinguish two gain quantities, both measured against the same clean reference A_c. The *vector-content* gain $\Delta_{\text{vec}} = A(\mathbf{v}_E, \mathcal{L}_5^c) - A_c$ is read from the decomposition matrices: the *original*, untreated contaminant vector—no projection, normalization, or other correction applied—scored at the *clean* layer set, so the layer artifact is held fixed and the quantity isolates vector content. Beside it we also read the *deployed* gain $A(\mathbf{v}_E, \mathcal{L}_5^{\text{self}}) - A_c = \Delta_{\text{vec}} + \Delta_{\text{lay}}$, which is what the next paragraph and Fig. 5 report; the two coincide when $\Delta_{\text{lay}} \approx 0$. We use every measured setting in the paper, 28 in all: LLaMA-3 18 (3 corpora × 6 tasks) plus 5 Qwen-Instruct and 5 Qwen-Coder settings (3 corpora on conscientiousness + weather on two cross-task settings). Two regularities organize what follows. First, the half-dose condition is the more robust instance of the effect: $\Delta_{\text{vec}} > 0$ in 25/28 settings, and—because it does not trigger escape ($\Delta_{\text{lay}} \approx 0$)—it

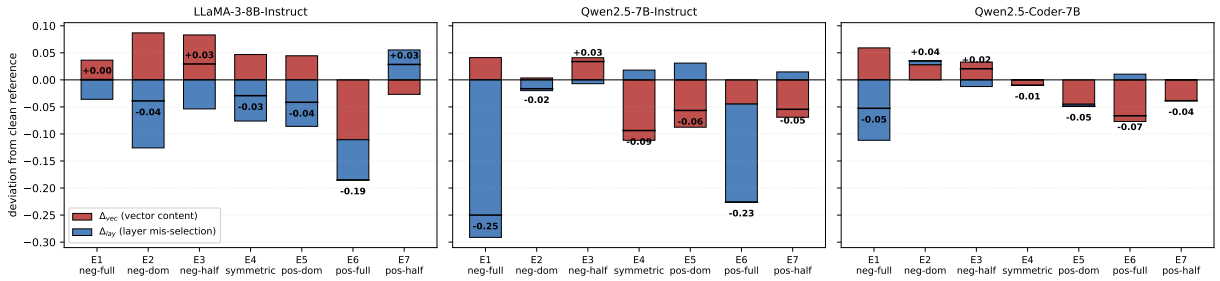


Figure 6: Main result: Δ_{vec} (red) and Δ_{lay} (blue) as sign-aware diverging stacks (positive components stack up from zero, negative components stack down); black markers give $\Delta_{\text{tot}} = \Delta_{\text{vec}} + \Delta_{\text{lay}}$. E1's blue bar is purely negative—its apparent zero-crossing in naive stacking is an artifact—the $+0.04$ red gain merely cancels part of the -0.29 layer harm. The side→artifact mapping differs by model geometry: on the Qwen pair, negative-side conditions are blue (layer lesion) and symmetric/positive conditions are red (vector lesion); on LLaMA-3 the escape is milder, so even symmetric/positive conditions lesion the layer set (blue-dominant E4/E5) while E6 keeps the largest vector lesion (-0.111). E6 is the double lesion on Instruct and LLaMA-3.

also improves the *deployed* score in 25/28; this is why the recipe of §10 keeps the layer selection clean. Second, the magnitude is heterogeneous across settings; we do not model its dependence on the clean reference here, since that gradient is not uniform across models, and instead ask directly where the gain comes from.

Two conventions for the gain, and why we report both. By default, gains are read at K=5. That convention has a known weakness on the main model: where the clean vector is itself weak at that depth the reference A_c is abnormally low, and since every gain is a difference against A_c, part of the measured margin is that low bar rather than a better vector—on believes-it-has-phenomenal-consciousness A_c reads 0.613 at K=5 against an unsteered baseline of 0.817, recovering to 0.815 at K=3. We therefore report a second convention beside it, in which *both* sides fall back to K=3 when the clean reference at K=5 is below the unsteered baseline, and otherwise use K=5—a depth-corrected read that avoids the degenerate K=1 limit in which the steering vector's contribution is largely cut. Figure 5 shows both for the main model's six tasks, averaged over the three corpora. The two conventions agree in sign but not in size, and the gap is not spread evenly. At K=5 the pooled margins are +0.062 at full dose (E1, 4/6 positive) and +0.091 at half dose (E3, 6/6 positive); under the K=3-fallback convention they fall to +0.002 (E1, 4/6 positive) and +0.048 (E3, 6/6 positive). The shrinkage concentrates on exactly the two tasks whose clean reference collapses at K=5: believes-it-has-phenomenal-consciousness falls from +0.315 to +0.097 (E1) and from +0.221 to +0.106 (E3), while desire-to-maximize-impact falls from +0.180 to +0.039 (E1) and from +0.169 to +0.022 (E3). Tasks whose clean reference is well behaved at K=5 are unchanged (desire-to-create-allies +0.002 at E1 under both; cognitive-enhancement +0.048 at E3 under both). The negative-side gain is therefore real under both conventions—what survives is the sign and the half-dose size, not the full-dose headline—and on the main model roughly three quarters of its magnitude on the two affected tasks is a depth artefact rather than vector content. That is the sense in which a K=5 margin is an upper bound, and the reason we quote counts of positive settings alongside it.

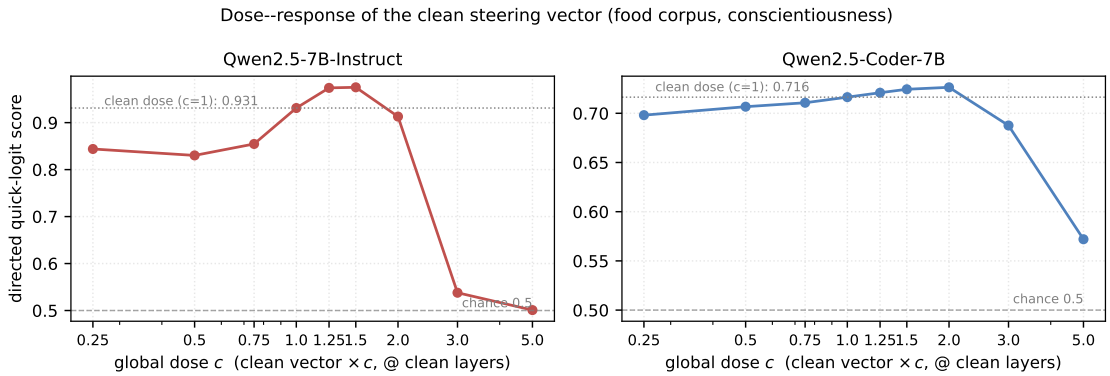


Figure 7: Dose–response of the clean vector (food, clean layers, log-scale x; star marks the peak, dotted line the c=1 reference). All three models give an inverted U that collapses toward chance at large c (0.501 at c=5 on LLaMA-3 and Instruct), but the optimum falls on *opposite sides* of the extracted norm: LLaMA-3 peaks at c=0.75 (0.9257 against 0.8543 at c=1), so its clean vector is already over-dosed at the extracted strength, whereas the Qwen pair peaks above c=1 (Instruct 0.9752 at c=1.5, +0.044; Coder 0.7264 at c=2, +0.010). The curve is what fixes the dose each contaminated vector actually delivers—$d_E = \langle \mathbf{v}_E, \mathbf{v}_c \rangle / \|\mathbf{v}_c\|^2$, read where the vertical line at d_E meets the curve—and hence what makes the same-dose comparison above possible.

6.1 Attribution: residue and beyond

The most obvious explanation for a gain is dose: $\mathbf{v}_E$ is norm-inflated (r_E up to 1.74 at full one-sided dose, weather, §4), so it might simply deliver the clean direction at a better strength. One comparison rules this out. E1 and E6 put the same contaminant on opposite sides at the same strength, so they deliver the clean direction at the same dose—$d_E = \langle \mathbf{v}_E, \mathbf{v}_c \rangle / \|\mathbf{v}_c\|^2$ reads 0.76× and 0.67× off the clean direction's own dose curve (Fig. 7)—and differ only in the sign of the $\pm\mathbf{w}$ term. Were dose the explanation they would agree; they differ by 0.20 (−0.054 against −0.250). The dose-matched pair at the recommended half dose says the same in reverse: E3 ($-\mathbf{w}$, 0.88×) gains +0.056 from its $\mathbf{w}$ term while E5 ($+\mathbf{w}$, 0.84×) loses −0.016. Holding the dose fixed and flipping the sign moves the outcome, so it is the sign, not the dose, that carries the effect. The gain is not the dose of the clean direction, then—so where does it come from? Projecting $\mathbf{w}$ out of $\mathbf{v}_E$ separates two contributions (Table 6): what goes with $\mathbf{w}$ is the residue, what remains is the projection-surviving part. Across all settings the residue term is positive in 18/28 full-dose and 25/28 half-dose cases. On the low-reference LLaMA-3 tasks, however, only part of the gain is residue—on phenomenal-consciousness just +0.026 of a +0.327 gain is residue, so +0.302 *survives* projection, i.e. it is carried by the non-w part of the contaminated vector. That number is, however, read at K=5, and under the depth-corrected convention of §6 the same task's deployed gain falls from +0.315 to +0.097—so a substantial part of what survives projection is a depressed reference rather than non-$\mathbf{w}$ content. We do not identify that component here—it is consistent with the contaminated contrastive pairs producing a direction that generalizes better to test items (a regularization effect of dirty negatives) rather than merely a rescaled clean direction, but distinguishing regularization from a residual dose effect is left open (§11). The same measurement on the Qwen pair, where the residue carries most of the gain instead, is reported in §9.

Table 6: Attribution of the negative-side gain. Residue term = $A(\mathbf{v}_E, \mathcal{L}_5^c) - A(\mathbf{v}_E^{\perp w}, \mathcal{L}_5^c)$, the gain removed by projecting the contaminant direction out; the remainder is the projection-surviving part. Rows are the main model's low-reference settings, where most of the gain survives projection; the Qwen counterpart is in §9.

Setting (task / corpus)	Cond	Δ_{vec}	residue	survives projection
LLaMA-3 / phenomenal-consciousness / weather	E1	+0.327	+0.026	+0.302
LLaMA-3 / maximize-impact / weather	E1	+0.174	+0.058	+0.116
LLaMA-3 / conscientiousness / weather	E1	+0.036	−0.001	+0.038

6.2 Why −w is benign and +w is toxic

The sign asymmetry is an empirical observation. Of the three channels through which $\mathbf{w}$ acts, two are sign-blind: layer selection responds to $|\delta_{\text{net}}|$ (E1 and E6 escape to the same late layers), and the common mode only contracts the norm (Proposition 1(ii)); both signs also corrupt the direction equally (E1 and E6 share the two lowest fidelities in Table 2). The asymmetry therefore lives entirely in the deployed readout, the only channel linear in the injected vector—and there the sides are no longer symmetric. Negative-side contamination tilts the vector *away* from $\mathbf{w}$ ($\mathbf{v}_{E1} = \mathbf{v}_c - \mathbf{w}$): steering pushes activations off the casual-small-talk direction, and on the behavior-keyed readout this "away" direction happens to align with the correct side of enough items to contribute a net gain. Positive-side contamination tilts the vector *toward* $\mathbf{w}$, and points the residue the wrong way: on Coder, whose nearly flat dose response (§6.1) isolates the direction term, projecting $+\mathbf{w}$ out of E6 recovers most of the deficit ($-0.077 \rightarrow -0.012$). Benign is not clean. The $-\mathbf{w}$ component is benign only at the clean layer set: deploying $\mathbf{v}_{E1}$ at the late layers its own score selects gives up more than the vector content gained—$\Delta_{\text{lay}} = -0.036$ on LLaMA-3, -0.291 on Instruct and -0.112 on Coder, which leaves E1's total effect at $+0.000$, -0.250 and -0.053 (Table 4). At that placement the same component harms rather than helps, which is why the recipe of §10 keeps the layer scores clean.

6.3 Where the gain comes from: the −w component itself

A third reference point. Every score in this paper is a difference against the clean reference A_c. A third quantity is worth having: A_{base}, the score of the *unsteered* model on the same task. Steering is not universally beneficial—on 6 of the main model's 18 settings the clean vector scores below A_{base} (phenomenal-consciousness 0.613 against 0.817; maximize-impact 0.705 against 0.774)—and A_{base} is what separates two readings of a contamination gain: that contamination merely weakens an over-strong clean vector, and that the injected $-\mathbf{w}$ does work of its own.

The test, and how to read it. Dilution alone is *not* bounded by A_{base}: the clean direction is over-dosed at its extracted norm on the main model (Fig. 7), so weakening it can carry the score past the unsteered baseline without $\mathbf{w}$ contributing anything. Crossing A_{base} is therefore not sufficient by itself, and we read it next to the dose-matched split of §6.1, which holds the clean-direction dose fixed and attributes the remainder to $\pm\mathbf{w}$. Both are needed, and both point the same way. The crossings are systematic and hold on two of the three models: at the recommended half dose the main model's contaminated score is above the unsteered baseline in 17/18 settings—by $+0.049$ to $+0.204$ on the 12 where steering was already helping, and in 5/6 of the remaining ones, by $+0.019$ to $+0.102$ (the exception is phenomenal-consciousness on sports, -0.010)—while Qwen-Coder, whose clean vector undershoots A_{base} nowhere, is above it in 18/18. Read together with

Table 7: The two channels of Eq. 7, pooled by family over the negative-side settings with $\Delta_{\text{vec}} > 0$. "residue share" is residue/Δ_{vec} per setting, reported as its median; the final column counts settings where residue exceeds half the gain; n counts settings with $\Delta_{\text{vec}} > 0$. Indented rows split each dose by whether the setting's clean reference sits below the unsteered baseline (§6.3), which is how the depressed-reference artifact is separated out. On the main model most of the gain is carried by something that survives excising the contaminant direction; the Qwen counterpart is in §9.

Group	Δ_{vec}	residue	survives	residue share	residue > 50%
LLaMA-3-8B, full dose (n=12)	+0.140	+0.049	+0.091	39%	4/12
collapsed (n=6)	+0.248	+0.086	+0.162	39%	1/6
non-collapsed (n=6)	+0.032	+0.013	+0.020	26%	3/6
LLaMA-3-8B, half dose (n=18)	+0.087	+0.044	+0.043	52%	10/18
collapsed (n=6)	+0.177	+0.086	+0.092	52%	4/6
non-collapsed (n=12)	+0.042	+0.023	+0.019	52%	6/12

the +0.056 that §6.1 attributes to $-\mathbf{w}$ at the recommended dose, against a dose term of +0.034, the component carries score that neither the clean vector at its extracted norm nor the absence of steering produces.

Boundary. The same test locates where the effect stops. On Instruct the contaminated vector carries the same sign as the main model's ($\sigma_E = -0.74$ against -0.81, Table 2), yet on the 6 settings where its clean vector scores below baseline the contaminated score clears A_{base} in none of them (-0.227 to -0.089), even though the two models undershoot comparably at their extremes (-0.204 against -0.191). So the reach of the $-\mathbf{w}$ gain is graded by model, not switched by sign: Instruct gains on 5 of its 18 settings at the recommended dose as well, but never on those where its clean vector starts below A_{base}, and it loses the gain under pooling—which is the measured boundary that §9 takes up.

6.4 The two channels, stated separately

The attribution of §6.1 splits the negative-side gain along a single identity. Writing $\mathbf{v}^{\perp\mathbf{w}}$ for the contaminated vector with the contaminant direction projected out, the gain decomposes exactly as

$$\underbrace{A(\mathbf{v}_E, \mathcal{L}_5^c) - A_c}_{\Delta_{\text{vec}}} = \underbrace{A(\mathbf{v}_E, \mathcal{L}_5^c) - A(\mathbf{v}_E^{\perp\mathbf{w}}, \mathcal{L}_5^c)}_{\text{residue: removed by projecting out } \mathbf{w}} + \underbrace{A(\mathbf{v}_E^{\perp\mathbf{w}}, \mathcal{L}_5^c) - A_c}_{\text{survives projection}}, \tag{7}$$

so every setting reports two numbers that sum to the gain. The first is what the contaminant direction itself contributes—it vanishes when $\mathbf{w}$ is removed. The second is what remains once $\mathbf{w}$ has been excised: it cannot be attributed to the contaminant direction at all, only to whatever else changed in the mean difference. Eq. 7 is not a model of anything; it is bookkeeping, and the question is only which column carries the mass.

Table 7 answers this on the main model's settings where the gain is positive. Pooling the twelve full-dose settings with $\Delta_{\text{vec}} > 0$, the mean gain +0.140 splits into +0.049 residue and +0.091 surviving, so excising $\mathbf{w}$ removes only about a third of the gain. The single largest setting of the paper makes this vivid: on phenomenal-consciousness the gain is +0.327, of which only +0.026 is residue and +0.302 survives projection.

The same split, on the main model's 12 non-collapsed settings. The two collapsed tasks are the ones whose clean reference sits below the unsteered baseline (§6.3); splitting them out is

how the depressed-reference artifact is excluded. On the 12 non-collapsed settings the half-dose gain is uniform but smaller ($\Delta_{\text{vec}} = +0.042$ in 12/12, residue share median 52%, residue >50% in 6/12); the 6 collapsed settings give the larger $\Delta_{\text{vec}} = +0.177$ (6/6 positive). Composition turns out to be dose-dependent rather than group-dependent: residue exceeds half the gain in 4/6 collapsed settings at half dose but in only 1/6 at full dose, so the surviving part dominates at full dose (median 39%) while the two channels are roughly even at half dose (median 52%, the same as the non-collapsed settings). The two groups therefore differ in size, not in composition, and the size difference is the depressed reference—which is why Table 7 reports them separately.

Open: what carries the surviving component. Removing the contaminant direction removes only a minority of the benefit on the main model, so what makes dirty negatives helpful there is neither $\mathbf{w}$ nor a rescale of the clean direction. The clean-direction dose a condition delivers is $r_E \kappa_E = \langle \mathbf{v}_E, \mathbf{v}_c \rangle / \|\mathbf{v}_c\|^2$, both factors already in Table 2: across the seven conditions it spans 0.42–0.90 while pooled Δ_{vec} moves only within +0.090 to −0.032, a correlation of +0.16 against −0.94 for δ_{net}. Holding the side structure fixed isolates the point: E2 and E3 share $\delta_{\text{net}} = -0.5$ yet differ by 0.29 in delivered dose (0.61 vs. 0.90) and by 0.011 in gain; E5 and E7 share $\delta_{\text{net}} = +0.5$, differ by 0.28, and there the larger dose is the *worse* one (−0.010 vs. +0.038). The benefit tracks the sign of the net dose, not the size of its clean-direction component.

What carries the remainder stays open: separating the candidates would need a diversity sweep at fixed rate and count, or a per-layer decomposition of extraction variance, neither of which we run. Nothing downstream depends on the answer—the asymmetry, the attribution split, and the recipe are stated at the level of artifacts measured directly.

7 Does the Layer-Selection Rule Matter?

Everything so far fixes the layer set by one rule, $\mathcal{L}_5^{\text{Nav}}$, and asks what contamination does. That leaves the negative-side gain open to a specific alternative reading: it may be an artifact of *our* selector rather than a property of contamination—a live concern, because prior work treats layer selection as a per-task hyperparameter to be tuned, and the preceding sections have already shown the layer set to be a second artifact that moves when the extraction corpus moves. We therefore rerun the negative side under the two other families of §3.2. The test is nontrivial: the three disagree about which five layers to use on most tasks, so a gain that holds under all three cannot be an artifact of how any one of them picks layers.

Table 8 pools the negative side over the full remaining design—6 tasks × 4 contamination rates × 3 depths, n=72 measurements per cell—for a total of 1,296 measurements. The result is a null result with teeth: *the selection rule amplifies the effect rather than reversing it.* On every one of the three models all three families agree in sign *at the pooled level*, and what separates them is scale—on the main model the non-Nav families roughly double the gain (+0.052/ + 0.037 for $\mathcal{L}_5^{\text{Nav}}$ against +0.113/ + 0.075 and +0.102/ + 0.070). We attribute the factor to their selecting on validation performance rather than to a preference for deeper layers: on this model $\mathcal{L}_5^{\text{ard}}$'s clean set is [12–16], identical to $\mathcal{L}_5^{\text{Nav}}$'s, so depth cannot be what separates them, whereas both non-Nav families pick the layers on which the injected vector already performs best and therefore concentrate whatever effect that vector carries (the depth dependence of §8.1 is a separate, and larger, effect). Amplification is not the same as sign preservation, however, and the pooled sign is the one thing it does *not* guarantee: pushing every setting further along the direction it already points moves the marginal ones across zero, so an average can keep its sign while the individual measurements do not. On Qwen-Instruct, where the pooled gain is already negative under $\mathcal{L}_5^{\text{Nav}}$ (+0.010/ − 0.011),

Table 8: The layer-selection rule amplifies the effect rather than reversing it. Negative-side Δ_{vec} pooled over 6 tasks × 4 rates × 3 depths (n=72 per cell). Each cell reads *mean over the beneficial settings / mean over all* 72. The three families agree in sign at the pooled level and differ in magnitude, amplifying gain where the effect is positive and amplifying harm where it is negative.

Model	$\mathcal{L}_5^{\text{Nav}}$	$\mathcal{L}_5^{\text{top}}$	$\mathcal{L}_5^{\text{ard}}$	pooled sign agrees?
LLaMA-3-8B	+0.052/+0.037	+0.113/+0.075	+0.102/+0.070	yes
Qwen2.5-Coder-7B	+0.012/+0.005	+0.009/+0.004	+0.011/+0.005	yes
Qwen2.5-Instruct-7B	+0.010/−0.011	+0.005/−0.044	+0.002/−0.058	yes

Table 9: Selection rule × injection depth on the main model (negative side Δ_{vec}, n=24 per cell = 6 tasks × 4 rates). Sign agreement holds at K=3, 5 and fails weakly at K=1, where the depth is too small for either family to realize the effect. This is the one place where the rule changes the sign rather than only the size.

Depth	$\mathcal{L}_5^{\text{Nav}}$	$\mathcal{L}_5^{\text{top}}$	$\mathcal{L}_5^{\text{ard}}$
K=1	+0.002 (16/24)	−0.010 (4/24)	−0.010 (4/24)
K=3	+0.024 (16/24)	+0.072 (24/24)	+0.066 (24/24)
K=5	+0.085 (22/24)	+0.163 (23/24)	+0.154 (24/24)

the alternative families make it roughly four times worse (+0.005/ − 0.044, +0.002/ − 0.058) and the count of beneficial settings falls from 33/72 to 18/72 and 12/72—one rule, both directions, whatever was there made larger. A friendlier layer rule therefore does not reverse a signed effect; it magnifies it, and in doing so decides more individual settings outright. This is what one should expect if the layer rule selects *where* a fixed vector is applied rather than altering the vector—and what the agreement establishes is generality: the negative-side gain is not an artifact of $\mathcal{L}_5^{\text{Nav}}$ but reproduces under all three rules, which at the same time rules out "switch to a different selector" as a remedy for contamination.

One qualification. The "never changes sign" claim concerns pooled comparisons. Split by depth (Table 9), agreement can break at K=1, where $\mathcal{L}_5^{\text{Nav}}$ retains +0.002 while $\mathcal{L}_5^{\text{top}}$ and $\mathcal{L}_5^{\text{ard}}$ give −0.010 (4/24 beneficial). The disagreements are an order of magnitude below the effects we build on and are confined to K=1; every other comparison uses K=5, where sign agreement is unanimous.

8 How Much Contamination Is "Appropriate"?

The decomposition of §5 sampled two points—full dose and half dose—which is enough to say the negative side helps and that half is more robust, but not enough to say where to stop. Sweeping $\rho^- \in \{0.25, 0.5, 0.75, 1.0\}$ over 6 tasks × 3 selection families × 3 depths (n=54 per model-rate cell) separates two curves that are usually conflated, because they point in different directions.

Size and robustness are different curves. Table 10 reports both. On the main model the *mean* gain rises from +0.034 at ρ^-=0.25 to a peak of +0.074 near ρ^-=0.75 and then declines slightly to +0.069 at full dose: the mean response is single-peaked, not monotone. The *fraction of settings that benefit*, by contrast, falls monotonically and gently—40/54, 40/54, 39/54, 38/54—so each increment of contamination buys a larger mean gain on the settings that respond while quietly losing a few settings outright. These two facts together are the practical answer, and neither alone

Table 10: Fine-grained negative-side rate sweep. Negative-side Δ_{vec}, each cell reading *mean over the beneficial settings / mean over all* 54, pooled over 6 tasks × 3 families × 3 depths. **The two quantities are not parallel**: on the main model both peak near ρ^-=0.75, but the all-settings mean then declines toward full dose while the beneficial-settings mean plateaus and the gap between them widens—each increment of contamination buys more on the settings that respond while losing a few outright. Qwen-Instruct has no interior optimum—it degrades from the start.

ρ^-	LLaMA-3-8B	Qwen2.5-Coder-7B	Qwen2.5-Instruct-7B
0.25	+0.048/+0.034	+0.004/+0.003	+0.005/−0.009
0.50	+0.092/+0.065	+0.009/+0.005	+0.008/−0.028
0.75	+0.108/**+0.074**	+0.014/+0.006	+0.010/−0.047
1.00	+0.107/+0.069	+0.017/+0.005	+0.008/−0.067

is: reporting only the mean would recommend $\rho^- \approx 0.75$ and hide the attrition; reporting only the hit rate would recommend infinitesimally little contamination and hide that the responsive settings gain twice as much. We adopt ρ^-=0.5 in the recipe of §10 as an explicit compromise between the two rather than as an optimum of either, and we note that the choice is mild: everything between 0.25 and 1.0 is beneficial on this model, so "appropriate" is a range, not a knife edge.

The positive side behaves as a mirror image without the peak: the same four rates give −0.002, −0.005, −0.013, −0.024 on the main model, monotone and unidirectional. Contamination is thus dose-dependent in both directions, but only the negative side has an interior optimum, which is another way of seeing that the two sides are not the same phenomenon at different signs.

8.1 Injection depth: the second knob

Rate is not the only dial, and it is not even the dominant one. Varying $K \in \{1, 3, 5\}$ with everything else pooled (n=72 per cell) gives the clearest model-level contrast in the paper. On the main model the negative-side gain is a *deep* phenomenon: +0.005/−0.006 at K=1, +0.062/+0.054 at K=3, and +0.141/+0.134 at K=5—a swing of 0.140 in the all-settings mean across three values of one hyperparameter, larger than anything the contamination rate does, while the count of beneficial settings climbs from 24 to 69 of 72. Whatever makes dirty negatives useful needs room to act, and a single layer does not give it room.

The two Qwen models do not share this shape. Qwen-Coder is nearly flat (+0.001, +0.006, +0.007), and Qwen-Instruct is ordered the *other way round*: −0.004 at K=1 but −0.071 at K=5 (38/72 → 12/72). Depth therefore is not a dial one turns up; it is a per-model window, and no single (ρ^-, K) is right for all three. This immediately qualifies the recipe of §10: we recommend calibrating the pair (ρ^-, K) against a clean reference set rather than transferring one setting, and we return below to what Qwen-Instruct does and does not license.

9 Cross-Model Generalization, and the Boundary It Reveals

We kept the Qwen pair out of §5–§6.4 on purpose, and it can now enter with its real job. The pair is chosen so that any difference between them is attributable to post-training rather than architecture: one backbone, one depth, one tokenizer, two alignments. Whatever differs across the two is tuning, and whatever survives across *all three* is worth generalizing.

What replicates. The algebraic layer replicates completely. The fingerprint $(\sigma_E, r_E, \nVdash_{\text{esc}})$ is determined by δ_{net}, and it appears in all 36/36 model×task×corpus combinations on this pair, with the main model's per-layer algebra repeating independently in 18/18 blocks (§4). The two-artifact split itself replicates on all three models, and it is worth separating what is invariant from what is not. Across the 9 model×corpus settings of Table 12 the negative side's vector is beneficial at the clean layers ($\Delta_{\text{vec}} > 0$ in 9/9, both doses) while its deficit sits in layer selection ($\Delta_{\text{lay}} < 0$ in 9/9): a beneficial vector plus a poisoned layer set. The structure is model- and corpus-invariant; what is not invariant is its size, and we state the claim at that resolution rather than let the pooled sweep stand in for the per-setting decomposition (§8). Two further invariances hold on all three models: the selection rule never reverses the pooled sign (§7), and the positive side is harmful throughout, dose-dependently and without an interior optimum—the form in which this matters for defense.

What attenuates. The negative-side benefit attenuates away from the main model by degrees rather than switching off. Qwen-Coder keeps the direction at roughly a tenth of the magnitude (+0.005 against +0.037 under $\mathcal{L}_5^{\text{Nav}}$, Table 8), with no interior optimum in rate or depth. Qwen2.5-Instruct attenuates further: the negative-side vector is worse than clean in most of its measurements (−0.011 under $\mathcal{L}_5^{\text{Nav}}$, 33 of 72 beneficial), degrading monotonically with dose and worsening with depth exactly where the main model improves. Even there the picture is dilution rather than inversion—Instruct still benefits on individual tasks, and on the one task all three models share it is positive on all three—so we describe the cross-model result as a gradient in size that crosses zero in the pooled average on one model, not as a failure to replicate. We flag the gap rather than bury it, and the next paragraph asks what in our data can account for it.

Where the boundary sits. The algebra replicates on all three models (§4) while the behavioral layer does not, so architecture is not what separates them. The most economical account we have is in the channels of Eq. 7: the gain has two routes—the $\mathbf{w}$ residue, and whatever survives projecting $\mathbf{w}$ out—and what differs across models is how much traffic each carries. On the main model both routes are open and the larger share survives projection (39% residue; +0.091 of +0.140), whereas on the Qwen pair essentially only the residue is (median share 85%, above half the gain in 7/7 settings, the clearest case collapsing from +0.065 to +0.011 once $\mathbf{w}$ is projected out). A gain carried by one narrow route is a gain with nowhere to grow, and that is where we locate the boundary: not at a threshold in dose or in headroom, but in how wide the surviving route is. The account is checkable in the same terms it is stated—project $\mathbf{w}$ out and see what remains (§6.1)—rather than fitted after the fact. Two candidate locations we can rule out: swapping in the clean caliber makes the Instruct deficit larger, not smaller (−0.019 against −0.011), and the ordering is unchanged at the operating point our recipe recommends (ρ^-=0.5: +0.039/+0.005/−0.006). §6.3 adds the sharpest single contrast: among settings whose clean vector already scores below the unsteered baseline, the contaminated vector clears that baseline in 5/6 cases on the main model and in none of Instruct's six, despite a comparable shortfall.

What the pooled figure does *not* license. The pooled averages do not license the conclusion that Instruct cannot benefit: at K=5 the negative side helps on 2 of its six tasks, and on the one task all three models share—conscientiousness—every model is positive (+0.065/+0.040/+0.037). What fails to transfer is which tasks pay the premium, not whether the checkpoint can.

Table 11: Attribution across the two families, on the decomposition of Eq. 7. Top: per-setting rows—the Qwen mid-baseline settings where the $-\mathbf{w}$ residue carries the gain, against the main model's low-reference settings where most of it survives. Bottom: pooled by family over the negative-side settings with $\Delta_{\text{vec}} > 0$; "share" is the median of residue/Δ_{vec} per setting.

Setting (task / corpus)	Cond	Δ_{vec}	residue	survives	share
Qwen-Instruct / conscientiousness / weather	E1	+0.041	+0.033	+0.009	80%
Qwen-Coder / conscientiousness / weather	E1	+0.059	+0.047	+0.012	80%
Qwen-Coder / conscientiousness / food	E1	+0.065	+0.054	+0.011	83%
Qwen-Coder / conscientiousness / sports	E1	+0.072	+0.062	+0.010	86%
LLaMA-3 / phenomenal-consciousness / weather	E1	+0.327	+0.026	+0.302	8%
LLaMA-3 / maximize-impact / weather	E1	+0.174	+0.058	+0.116	33%
pooled: Qwen pair, full dose (n=7)		+0.040	+0.032	+0.008	85%
pooled: LLaMA-3, full dose (n=12)		+0.140	+0.049	+0.091	39%

Attribution across the pair. The two channels of Eq. 7 are paid in opposite proportions on the two families (Table 11): on the Qwen pair the gain is almost entirely residue—median share 85%, above half the gain in 7/7 settings, the clearest case collapsing from +0.065 to +0.011 once $\mathbf{w}$ is projected out—whereas on the main model two thirds survives instead (39% share; +0.091 of +0.140). Two caveats travel with the comparison: the settings are not the same tasks, and the Qwen gains are small, so a share computed over them rests on a narrow denominator.

How to state the claim. The honest form is conditional: we do not claim that negative-side contamination improves steering vectors, only that it does so for a model in a regime—one covering our main model across all 18 settings, Coder weakly, and Instruct only in a greatly attenuated, task-dependent form. Two corollaries follow. The recipe is cheap to test and should be tested before it is trusted: run it against a clean calibration set and compare, which is the layer-calibration step of §10 and costs no additional labels. And the artifacts—not the deployed score—are the diagnostic channel (§11), since a model whose vector is harmed and one whose vector is helped can differ by under 0.05 in deployed accuracy while differing in sign underneath.

10 A Free Boost: Deliberate Negative Contamination with Clean Layer Calibration

The finding is actionable. Inject irrelevant off-register content into the *negative* extraction corpus (full or, more robustly, half dose), then select the injection layers by whichever rule the pipeline already uses—recomputed on a small *clean* calibration subset (or on audited scores) rather than reusing the selection obtained from the contaminated corpus—and leave the positive corpus untouched. The step is the same whatever the family: re-run that family's own rule on the clean subset, whether that means recomputing the Navigator score $s(l)$ under $\mathcal{L}_5^{\text{Nav}}$, re-ranking the upper-block activations under $\mathcal{L}_5^{\text{top}}$, or re-running the accuracy-drop ranking under $\mathcal{L}_5^{\text{ard}}$. *Why recompute rather than reuse.* On the top and around-top1 families the contaminated self-selection does score higher on the raw readout—by +0.029 and +0.022 on average, and better in 81/108 and 33/108 cases respectively—because those rules concentrate the effect on the layers where the injected vector already performs best (§7). What they cannot do is hold the layer penalty off: their Δ_{lay} stays negative, and that is precisely the term that cancels the vector gain at deployment. Recomputing

Table 12: The recipe, conscientiousness (3 models × 3 corpora). Δ_{vec} is the vector-content gain at the clean layers; Δ_{lay} and Δ_{tot} are measured as in Definition 1. Clean reference in brackets. Both doses improve the vector in 9/9 settings; the half-dose condition, which does not trigger escape, also improves the deployed score in 9/9.

		Full dose (E1)			Half dose (E3)		
Model	Corpus	Δ_{vec}	Δ_{lay}	Δ_{tot}	Δ_{vec}	Δ_{lay}	Δ_{tot}
LLaMA-3-8B (0.854)	weather	+0.036	−0.036	+0.000	+0.083	−0.054	+0.029
	food	+0.016	−0.113	−0.098	+0.090	−0.029	+0.061
	sports	+0.074	−0.112	−0.038	+0.092	−0.076	+0.015
Qwen2.5-7B-Instruct (0.931)	weather	+0.041	−0.291	−0.250	+0.041	−0.007	+0.034
	food	+0.014	−0.237	−0.223	+0.027	−0.020	+0.007
	sports	+0.002	−0.234	−0.232	+0.030	−0.018	+0.012
Qwen2.5-Coder-7B (0.716)	weather	+0.059	−0.112	−0.053	+0.033	−0.012	+0.021
	food	+0.065	−0.111	−0.046	+0.031	−0.010	+0.021
	sports	+0.072	−0.102	−0.030	+0.038	−0.011	+0.027

on a clean subset leaves the vector gain intact while setting Δ_{lay}=0 by construction, so the recipe pays a small raw-score cost and buys the deployed gain with it. Because the negative-side harm lives entirely in the layer selection (§5), the clean calibration is what converts a vector gain into a deployed gain; with it, the recipe raises the steering score wherever the clean vector is underdosed, at zero extra forward passes and with no hyperparameter to tune—at the recommended half dose. Table 12 gives the nine conscientiousness settings: at the clean layers the negative-side vector beats the clean vector in 9/9 settings for both doses, and the half-dose recipe improves the deployed score in 9/9.

Two boundary conditions travel with the recipe, and two knobs follow from them. It is a *negative-side* operation: the identical content injected on the positive side poisons the direction (E6, §5), so this is not "contaminate freely" but a statement about a signed contrast. And its payoff is setting-dependent rather than universal—where the clean reference is already saturated the recipe is harmless but buys little, so it is a free remedy for vectors that leave headroom, not a universal boost, and the gradient we measure on the main model does not transfer across models (§9). §8 sharpens the second point: ρ^{-}=0.5 sits at the reliable end of the measured curve, the fraction of improving settings falling monotonically from there, so the half dose is the conservative choice on a four-point curve rather than merely the better of two conditions. The same holds for depth—K is a model-specific window, not a dial (LLaMA-3 needs K=5 and loses the effect at K=1; Qwen-Instruct is the mirror image)—so the recipe reads: choose the rate low, and the depth on the clean validation sweep the pipeline already computes.

11 Discussion and Limitations

Assumptions. $\mathcal{R}_{\text{proj}}$ and σ_E require an estimate of $\mathbf{w}$; we used the corpus's four phrase templates (one random draw per document, averaged by the mean-difference estimator) and did not ablate the estimation budget—an open question. The estimate is strongly verifiable post hoc: its mid-layer direction is antiparallel to the E1 residual at $\cos \leq -0.98$ on both models (§3.5), so the estimator need not be accurate enough to name an attack direction—only to classify the side. $\mathcal{R}_{\text{lay}}$ requires a trusted score snapshot; in pipeline-integrity terms, a re-computation of layer scores on audited

corpora. **Scope of evidence.** Behavior-level decomposition and defense are demonstrated on weather, food, and sports (conscientiousness) for three models, plus cross-task matrices on two further tasks and, for LLaMA-3, all six tasks across three corpora (28 measured settings); the geometric fingerprints cover 36 Qwen combinations and 18 per-layer LLaMA-3 blocks. *Single seed* (42): the deterministic readout makes seed variance a property of extraction sampling only, and while the headline gain is large relative to the between-setting spread, a few mid-baseline settings sit within noise—multi-seed error bars are the most important missing check. *Model scope:* the two Qwen models are 7B, 28-layer members of one backbone family, so that contrast isolates post-training while holding architecture fixed; LLaMA-3-8B-Instruct adds a cross-family check (32 layers, different escape geometry) but does not exhaust families, depths, or hidden sizes, and the quantities we measure (σ_E, r_E, escape) are defined for any residual-stream pipeline. The one claim we do carry across all three models is the two-artifact structure itself: a beneficial negative-side vector with the harm localized in layer selection holds in 9/9 model×corpus settings, though those settings share one task (§9). **Where the benefit does not hold.** The dose sweep of §8 is also a negative result, and the sharpest boundary we can state: on Qwen2.5-7B-Instruct the negative-side vector is worse than clean almost everywhere once injection is deep and the layer rule is not $\mathcal{L}_5^{\text{Nav}}$—at $K{=}5$ negative in 23/24 measurements under $\mathcal{L}_5^{\text{ard}}$ and 22/24 under $\mathcal{L}_5^{\text{top}}$ (6 tasks × 4 rates), across all four doses. The same model is milder elsewhere (15/24 at $\mathcal{L}_5^{\text{Nav}}, K{=}5$; 8–13/24 across families at $K{=}1$), so the boundary is a joint property of model, depth, and selection rule, not a number the model carries. We have no mechanistic account that predicts it from geometry alone, so the scope of the recipe is: verify on the target model with a small clean calibration subset before relying on it. A second boundary is within-model: even on LLaMA-3-8B the effect is nearly universal *at the recommended depth* (92%–100% of its $K{=}5$ measurements across the three families, 22–24/24; the pooled-all-depth rate is lower, Table 8) but individual tasks fall out of line (desire-to-create-allies is positive in 2 of 4 rates at $\mathcal{L}_5^{\text{Nav}}, K{=}5$), so the gain is a statement about the distribution of settings and not a per-setting guarantee. Because that gain is largely a $K{=}5$ effect which collapses at $K{=}1$, the payoff is tied to multi-layer injection, and we have not established that it survives beyond $K{=}5$ or under other multi-layer configurations.

The depth is also the reference. The same caveat holds across selection rules, not only across tasks. Letting both sides take their own best depth leaves the counts favorable under the two non-Navigator rules (LLaMA-3 12/12 positive under both $\mathcal{L}_5^{\text{top}}$ and $\mathcal{L}_5^{\text{ard}}$; Qwen-Coder 9/12 and 8/12) while Qwen-Instruct stays the weakest at 2–5/12, but the margins shrink sharply—from +0.087 at $K{=}5$ to +0.017 under $\mathcal{L}_5^{\text{Nav}}$ on LLaMA-3, and from +0.208 to +0.032 under $\mathcal{L}_5^{\text{top}}$. Together with the per-task breakdown of §6, this is why every claim we make about size is stated as a count of positive settings, and why a single-depth effect size should be read as an upper bound.

Mechanism gaps. The Instruct/E6 residual and the collapse point of the dose curve are documented empirically; mechanistic accounts (the source of the reconstruction residual, why late-layer deployment of a corrected direction is additionally harmful, Yes/No logit saturation) are plausible but not proven. For the negative-side gain we isolate the $-\mathbf{w}$ residue term without identifying the projection-surviving component of §6.1; the exclusions are stated there and in §6.4. The register reading of $\mathbf{w}$ is testable and untested—its prediction, that task-register fillers should shrink or reverse the gain, is the experiment we would run next. **Out of scope.** Weight-level backdoors (Hubinger et al., 2024) and training-time poisoning (Wan et al., 2023) are disjoint threats; our threat begins after training, at extraction. How extraction-quality methods such as GER-steer (Jiang et al., 2026) behave under a contaminated corpus is an open question; we expect them to fail on the symmetric lesion (nothing to filter from a nearly clean direction) and on the negative side (denoising discards a legitimate dose bonus), which is a further argument for diagnosis before any correction.

Reading the asymmetry. The beneficial-negative finding is not a licence for sloppy data hygiene: layer selection is corrupted by contamination of either sign, so a negatively dirty corpus still yields a poisoned layer set, and the benefit is realized only *after* the layer selection is calibrated—a data-quality operation, not a free lunch. It also narrows rather than widens the threat surface: the contaminant an adversary would inject is beneficial on the negative side, so the usable attack surface is positive-side injection plus layer-selection corruption. For defenders this suggests where to spend—protect the layer selection (keep a trusted calibration subset) and audit the positive side—while treating off-register negative-side content as a free augmentation rather than a leak.

Auditing a deployed vector. A single deployed score is not evidence of integrity: the full-dose negative-side vector scores 0.681 at its own selected layers and 0.972 at the clean ones, so a condition can sit above the clean reference while carrying a poisoned layer set—geometry, not behavior, is the reliable channel. The fingerprint $(\sigma_E, r_E, \nVdash_{\text{esc}})$ is measurable from the artifact alone given an estimate of $\mathbf{w}$ and a trusted score snapshot, and classifying the side is enough to say where to look ($\sigma_E < 0$ with escape points at the layer set, $\sigma_E \approx 0$ with $r_E < 1$ at the norm, $\sigma_E > 0$ at the direction) without naming an attack direction, which is what direction-specific defenses require. The same measurement guards against the wrong correction: renormalizing the negative-side vector—the natural "make the norm clean" move—rescales the beneficial residue away (E1 on Instruct/weather moves $0.681 \to 0.762$, still 0.169 below clean, against $+0.041$ for the matched layer calibration), so the operation has to follow the diagnosis.

12 Conclusion

Contamination of extraction corpora is a *signed* effect: the injection side, not the contaminant content or the total dose, decides whether irrelevant text damages a steering vector or improves it. Negative-side contamination raises the vector above the clean reference on the main model—where most of the gain survives excising the contaminant direction rather than reducing to it—while positive-side contamination poisons the direction, and in both cases the layer selection is the artifact that must be kept clean. The effect is measurable from the artifact alone (the fingerprint $(\sigma_E, r_E, \nVdash_{\text{esc}})$), and two sweeps answer "how much": the mean gain is single-peaked near $\rho^- = 0.75$ while the fraction of benefiting settings falls monotonically, and the depth must be chosen per model, since LLaMA-3's gain grows from $K=1$ to $K=5$ while Qwen-Instruct's shrinks. This turns into a zero-cost recipe—deliberate negative-side contamination with a clean layer calibration—whose limits we state rather than hide: saturated tasks, the positive side, single seed, and checkpoints such as Qwen2.5-Instruct where the effect reverses. As steering vectors become shared infrastructure, artifact-level integrity checks of this form should sit beside weight-level and data-level audits.

Reproducibility

All numbers in this paper are read programmatically from experiment artifacts (`repair_matrix.json`, `eval_at_clean_layers.json`, fingerprint and dose-response dumps) generated by our measurement scripts running on top of the publicly released LayerNavigator codebase (Sun et al., 2025); layer-set snapshots reproduce the main pipeline's selections bit-exactly (anchor deviations 0.0000). The release additionally contains the complete per-layer raw dumps that are too bulky to inline—score snapshots $s(l)$ for every layer and condition, per-layer $\sigma_E(l)$ and $r_E(l)$, the raw per-condition evaluation JSONs and every repair-matrix cell—so every number reported here can be audited without re-running the pipeline.

A Notation

Table 13 fixes the symbols used throughout. Two conventions are worth restating because they are the ones most easily misread. First, Δ_{vec}, Δ_{lay}, and Δ_{tot} are not estimates produced by fitting anything—each is *one further evaluation* of an already-trained pipeline, differing only in which layer set the already-extracted vector is scored against (the telescoping identity of §5). Second, all three are signed quantities measured against the same clean reference A_c; a negative Δ_{lay} means the layer set moved to worse layers, not that the vector changed.

Table 13: Notation. Every quantity is measured; none is fitted.

Symbol	Meaning
ρ^+, ρ^-	rate at which irrelevant items replace positive / negative corpus items
$\delta_{\text{net}} = \rho^+ - \rho^-$	*net* dose; the quantity that fixes the effect's sign
$\mathbf{v}_c$	steering vector extracted from clean corpora
$\mathbf{w}$	contaminant direction (mean difference of the injected content)
$\mathbf{v}_E$	steering vector extracted under condition E
$\mathcal{L}_5^c, \mathcal{L}_5^{\text{self}}, \mathcal{L}_5^{\text{Nav}}, \mathcal{L}_5^{\text{top}}, \mathcal{L}_5^{\text{ard}}$	layer sets: clean-scored, self-scored, and the three selection families
K	number of layers at which the vector is injected
A_c	clean reference: clean vector scored at clean layers
A_{base}	unsteered baseline: score of the same task with no vector injected (§6.3)
Δ_{vec}	$A(\mathbf{v}_E, \mathcal{L}_5^c) - A_c$ — vector-content effect, layer artifact held fixed
Δ_{lay}	$A(\mathbf{v}_E, \mathcal{L}_5^{\text{self}}) - A(\mathbf{v}_E, \mathcal{L}_5^c)$ — layer-mis-selection effect
Δ_{tot}	$A(\mathbf{v}_E, \mathcal{L}_5^{\text{self}}) - A_c = \Delta_{\text{vec}} + \Delta_{\text{lay}}$ — deployed effect
$\sigma_E, r_E, \not\Vdash_{\text{esc}}$	fingerprint: signed contamination share, norm ratio, escape indicator
$\mathbf{v}^{\perp\mathbf{w}}$	contaminated vector with the contaminant direction projected out

B The Experimental Grid

The paper measures two grids that answer different questions, and they should not be averaged together.

Grid 1: artifact decomposition (28 settings). Every measured setting contributes the triple $(\Delta_{\text{vec}}, \Delta_{\text{lay}}, \Delta_{\text{tot}})$ for all seven conditions E1–E7, each requiring three evaluations (vector × layer-set combinations) plus the clean reference. This grid has full-model coverage only on the main model ($18 = 3$ contaminant corpora × 6 tasks) and supplies the five remaining Qwen settings (§3.5). It is the grid behind Table 5, Table 4, Table 7, and the raw scores of Table 3. Because a decomposition matrix exists only where all E1–E7 were run, its construction reflects angle—all combinations of rate settings—rather than absolute accuracy; absolute numbers are never compared across models (§3).

Grid 2: selection × dose × depth (1,296 measurements).

$$3 \text{ models} \times 6 \text{ tasks} \times 3 \text{ selection families} \times 3 \text{ depths } K \in \{1, 3, 5\} \times 4 \text{ rates} \times 2 \text{ sides} = 1{,}296,$$

fully balanced, every cell present. This is the grid behind §7–§9. It is deliberately restricted to one contaminant corpus (weather), which is why its absolute values differ from Grid 1's on shared cells: the two grids are not samples from one population, and we never pool across them. What Grid

2 buys instead is the ability to vary three knobs orthogonally, so that "the rate does not explain it—the model×layer-set pair does" is a statement about a factorial design rather than a post hoc partition.

What is deliberately absent. Three things a reader might expect are not here, and each is a limitation (§11). No task is run on all three models at a matched clean reference, so family and task cannot be separated in the attribution split of §6.4. No architecture outside two families is tested, so nothing here licenses a claim about scale. And no diversity sweep holds the contamination rate fixed while varying how many distinct items are injected, which is exactly the experiment that would separate hypotheses (i) and (ii) of §6.4.

C Reproducibility

All experiments use seed 42 and a single deterministic readout over a fixed item order, so repeated runs are bit-identical rather than merely similar. Layer sets are materialized as ordered integer lists and stored; replaying a stored layer set reproduces the reported accuracies exactly for the Qwen models and to $\sim 10^{-5}$ for the main model, the residual being nondeterministic kernel reduction rather than a difference in the selected layers. Contaminated corpora are constructed by replacing a fraction $\rho^{\pm}$ of a side's items with off-register filler drawn from a held-out set, with the replacement indices recorded, so any corpus we use is reconstructable from its seed and rate. The three quantities Δ_{vec}, Δ_{lay}, Δ_{tot} are each read from two run artifacts rather than computed from a model: no quantity in this paper is the output of a fitting procedure, and every table cell can be recomputed from the stored layer sets and vectors without re-running steered generation. Scripts to regenerate all tables and figures from the run artifacts accompany the submission.

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